# Data Types And Structures Questions

1. What are the data structures, and why are they important?

- Data structures are ways to organise and store data in a computer. They are important because they help in efficient data storage, retrieval, and processing.

2. Explain the difference between mutable and immutable data types with examples.

- Mutable data types can be changed after creation.
- Examples: List, Dictionary, Set
- Immutable data types cannot be changed after creation.
- Examples: String, Tuple, Integer

3. What are the main differences between lists and tuples in Python?

- Lists are mutable, while tuples are immutable.
- Lists use [], tuples use ().
- Lists are slower compared to tuples.

4. Describe how dictionaries store data.

- Dictionaries store data in key-value pairs.
- Each key is unique and used to access its value.

5. Why might you use a set instead of a list in Python?

- Sets automatically remove duplicate values
- and provide faster searching.

6. What is a string in Python, and how is it different from a list?

- A string is a sequence of characters and is immutable.
- A list can store multiple data types and is mutable.

7. How do tuples ensure data integrity in Python?

- Tuples are immutable, so their values cannot be changed accidentally.

8. What is a hash table, and how does it relate to dictionaries in Python?

- A hash table stores key-value pairs efficiently.
- Python dictionaries are implemented using hash tables.

9. Can lists contain different data types in Python?

- Yes, lists can contain different data types
- such as integers, strings, and floats.

10. Explain why strings are immutable in Python.

- Strings are immutable to improve performance,
- security, and memory efficiency.

11. What advantages do dictionaries offer over lists for certain tasks?

- Dictionaries provide faster data access using keys.

12. Describe a scenario where using a tuple would be preferable over a list.

- Tuples are preferred when data should remain constant,
- such as coordinates (x, y).

13. How do sets handle duplicate values in Python?

- Sets automatically remove duplicate values.

14. How does the "in" keyword work differently for lists and dictionaries?

- In lists, "in" checks values.
- In dictionaries, "in" checks keys.

15. Can you modify the elements of a tuple? Explain why or why not.

- No, tuple elements cannot be modified
- because tuples are immutable.

16. What is a nested dictionary, and give an example of its use case?

- A nested dictionary is a dictionary inside another dictionary.

students = { "Rahul": {"age": 20, "marks": 85} }

print(students)

17. Describe the time complexity of accessing elements in a dictionary.

- Accessing elements in a dictionary usually takes O(1) time.

18. In what situations are lists preferred over dictionaries?

- Lists are preferred when ordered data storage
- and indexing are required.

19. Why are dictionaries considered unordered, and how does that affect data retrieval?

- Dictionaries retrieve data using keys instead of positions.

20. Explain the difference between a list and a dictionary in terms of data retrieval

- Lists use indexes for retrieval.
- Dictionaries use keys for retrieval.

## Practical Questions

```
#1. Create a string with your name and print it.
name = "Bandana"
print(name)
```

```
Bandana
```

```
#2. Find the length of the string "Hello World".
text = "Hello World"
print(len(text))
```

```
11
```

```
#3. Slice the first 3 characters from the string "python programming".
text = "python programming"
print(text[0:3])
```

```
pyt
```

```
#4. Convert the string "hello" to uppercase.
text = "hello"
print(text.upper())
```

```
HELLO
```

```
#5.Replace the word "apple" with "orange" in the string.
text = "I like apples"
print(text.replace("apples", "orange"))
```

```
I like orange
```

```
#6.Create a list with numbers 1 to 5 and print it.
list = [1,2,3,4,5]
print(list)
```

```
[1, 2, 3, 4, 5]
```

```
#7.Append the number 10 to the list [1,2,3,4].
numbers = [1,2,3,4]
numbers.append(10)
print(numbers)
```

```
[1, 2, 3, 4, 10]
```

```
#8.Remove the numbers 3 from the list [1,2,3,4,5]
numbers = [1,2,3,4,5]
numbers.remove(3)
print(numbers)
```

```
[1, 2, 4, 5]
```

```
#9.Access the second element in the list ['a','b','c','d'].
text = ['a','b','c','d']
print(text[1])
```

```
b
```

```
#10.Reverse the list [10,20,30,40,50].
numbers = [10,20,30,40,50]
print(numbers[::-1])
```

```
[50, 40, 30, 20, 10]
```

```
#11.Write a code to create a tuple with the elements 100,200,300 and print it.
numbers = (100,200,300)
print(numbers)
```

```
(100, 200, 300)
```

```
#12.Write a code to access the second to last element of the tuple('red','green','blue','yellow').
colors = ('red','green','blue','yellow')
print(colors[-2])
```

```
blue
```

```
#13.Write a code to find the minimum number in the tuple(10,20,5,15).
numbers = (10,20,5,15)
print(min(numbers))
```

```
5
```

```
#14.Write a code to find the index of the element "cat" in the tuple('dog','cat','rabbit').
animals = ('dog','cat','rabbit')
print(animals.index("cat"))
```

```
1
```

```
#15. Write a code to create a tuple containing three different fruits and check if "kiwi" is in it.
fruits = ("apple", "banana", "orange")
print("kiwi" in fruits)
```

```
False
```

```
#16. Write a code to create a set with the elements 'a', 'b', 'c' and print it.
letters = {'a', 'b', 'c'}
print(letters)
```

```
{'a', 'c', 'b'}
```

```
#17.Write a code to clear all elements from the set {1, 2, 3, 4, 5}.
numbers = {1, 2, 3, 4, 5}
numbers.clear()
print(numbers)
```

```
set()
```

```
#18. Write a code to remove the element 4 from the set {1, 2, 3, 4}.
numbers = {1, 2, 3, 4}
numbers.remove(4)
print(numbers)
```

```
{1, 2, 3}
```

```
#19. Write a code to find the union of two sets {1, 2, 3} and {3, 4, 5}.
set1 = {1, 2, 3}
```

```
set2 = {3, 4, 5}
print(set1.union(set2))
```

```
{1, 2, 3, 4, 5}
```

```
#20. Write a code to find the intersection of two sets {1, 2, 3} and {2, 3, 4}.
set1 = {1, 2, 3}
set2 = {2, 3, 4}
print(set1.intersection(set2))
```

```
{2, 3}
```

```
#21. Write a code to create a dictionary with the keys "name", "age", and "city", and print it.
person = {
    "name": "Bandana",
    "age": 20,
    "city": "Delhi"
}
print(person)
```

```
{'name': 'Bandana', 'age': 20, 'city': 'Delhi'}
```

```
#22. Write a code to add a new key-value pair "country": "USA" to the dictionary {'name': 'John', 'age': 25}.
person = {'name': 'John', 'age': 25}
person["country"] = "USA"
print(person)
```

```
{'name': 'John', 'age': 25, 'country': 'USA'}
```

```
#23. Write a code to access the value associated with the key "name" in the dictionary {'name': 'Alice', 'age': 30}.
person = {'name': 'Alice', 'age': 30}
print(person["name"])
```

```
Alice
```

```
#24. Write a code to remove the key "age" from the dictionary {'name': 'Bob', 'age': 22, 'city': 'New York'}.
person = {'name': 'Bob', 'age': 22, 'city': 'New York'}
del person["age"]
print(person)
```

```
{'name': 'Bob', 'city': 'New York'}
```

```
#25. Write a code to check if the key "city" exists in the dictionary {'name': 'Alice', 'city': 'Paris'}.
person = {'name': 'Alice', 'city': 'Paris'}
print("city" in person)
```

```
True
```

```
#26. Write a code to create a list, a tuple, and a dictionary, and print them all.
list = [1,2,3,4]
print(list)
```

```
[1, 2, 3, 4]
```

```
#27. Write a code to create a list of 5 random numbers between 1 and 100, sort it in ascending order, and print the result.
import random
numbers = [random.randint(1, 100) for _ in range(5)]
numbers.sort()
print(numbers)
```

```
[47, 84, 87, 94, 97]
```

```
#28. Write a code to create a list with strings and print the element at the third index.
fruits = ["apple","banana","mango","grapes","orange"]
print(fruits[3])
```

```
grapes
```

```
#29. Write a code to combine two dictionaries into one and print the result.
dict1 = {"a": 1, "b": 2}
dict2 = {"c": 3, "d": 4}
dict3 = {**dict1, **dict2}
print(dict3)
```

```
{'a': 1, 'b': 2, 'c': 3, 'd': 4}
```

```
#30. Write a code to convert a list of strings into a set.
fruits = ["apple", "banana", "orange"]
fruits_set = set(fruits)
print(fruits_set)
```

```
{'banana', 'orange', 'apple'}
```